

Wei-Ting Liu

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EDUCATION

2024 – Present **National Taiwan University** — B.B.A., Information Management
Double Major, Trans-disciplinary Bachelor Degree (College of Innovation)
Leadership Program

WORK EXPERIENCE

ABConvert — Reinforcement Learning R&D Intern 2026.07 – Present
– Designed a **dual-loop RL system** that trains agents to optimize Shopify stores, aiming to build an accurate, real-world-calibrated simulator that closes the sim-to-real gap.

NTU INSIGHT — AI Intern 2026.02 – 2026.07
– Developed an **internal service-design product** and led **AI adoption workshops**, moving 80% of staff from chat-based AI use to building their own website products.

ABConvert — Software Engineer Intern 2026.02 – 2026.06
– Migrated **abconvert.io** (10 pages) from Webflow to a self-hosted Next.js app, reaching a 97 Lighthouse SEO score and enabling custom interactive components.
– Built **internal dashboard** features (experiment research, visualization, analysis) from a cross-team needs survey, eliminating ~70% of manual metric lookups for Customer Success and the CEO.

ABConvert — Startup Generalist Summer Intern (Product/AI) 2025.07 – 2025.09
– **AI-native generalist**: LLM-assisted workflow to compress spec → implementation loop; shipped Next.js + TypeScript frontend (SSR/ISR, streaming UI) at startup pace.

RESEARCH EXPERIENCE

Multi-Agent RL for Decentralized Epidemic Surveillance | [10.5281/zenodo.21619163](https://doi.org/10.5281/zenodo.21619163) 2026.02 – Present
– Built a **CTDE multi-agent RL framework** (MAPPO/QMIX/IPPO/CQL on PettingZoo) modeling Brazil's 26 states as agents alerting on dengue from 258 weeks of case counts and Google Trends, benchmarked against an information-parity threshold rule.
– Showed the agents lose systematically on historical replay and **decomposed the gap** into reward misalignment (partly recovered by a gated false-alarm penalty) and residual on-policy sample inefficiency; the same agents do learn control on action-coupled SIR/SIS environments, localizing the failure to task specification rather than to RL. Released a 107-checkpoint replication archive; manuscript in preparation.

Age-Mixing Structure in Aggregated Urban Mobility Data | [10.5281/zenodo.22152089](https://doi.org/10.5281/zenodo.22152089) 2026.08 – Present
– Measured **age-assortative mixing across 10.2 billion rows** of Seoul's Living Migration release (79 months, 424 administrative dong, 16 age bands), pairing the near-zero reading with a synthetic-world calibration that fixes the instrument's detection limit.
– Found the co-arrival matrix within **0.00202 of proportionate mixing** in every month — a factor of 33–35 below a national contact survey — while the pipeline recovers 92–101% of true assortativity on synthetic controls; verified by a mutation-tested citation audit, a determinism harness, and an independent re-implementation agreeing on 436/436 checks.

PROJECTS

Spotify GuessSong [GitHub](#) | [Live](#)
Next.js, TypeScript, Spotify Web API. Real-time multiplayer music guessing game with 70,000+ monthly active users; host pastes any Spotify playlist URL and the app runs a round-by-round quiz with live scoring.

SKILLS

Stack	TypeScript, React, Next.js, PostgreSQL, Tailwind, Python, PyTorch, C++
Tools	REST/GraphQL, Git; Service Design, Design Thinking
Coursework	Data Structures, Web Programming, Algorithms, Operating Systems

AWARDS & LEADERSHIP

2025 **PDAO Competition** (Programming Competition) — 5th overall, 1st in IM Department.
2026.07 – 2027.06 **NTU Information Management Student Association** — Vice President.